

SERVER ENVIRONMENT AND CONTROL SYSTEM FOR CSPC-ICTU:

bucket: monitoring

org: capstone

1. Source - ESP32, Interval - 60s, 4 fields

Measurement: sensor_environment

Tags: device_id, device_name, location, source("live"/"sd_backup"), smoke_status(string), environment_status(string)

Fields: temperature(float), humidity(float), smoke_ppm(float), heat_index(float)

2. Source - Go Agent, Interval - 30s, 11 fields

Measurement: server_metrics

Tags: device_id, device_name, location, os

Fields: cpu_percent(float), mem_used_mb(float), mem_total_mb(float), mem_percent(float), disk_used_gb(float), disk_total_gb(float), disk_percent(float), net_bytes_sent(float), net_bytes_recv(float), uptime_seconds(float), process_count(integer)

3. Source - RouterOS API, Interval - 60s, 8 fields

Measurement: network_traffic

Tags: device_id, device_name, interface_name, location_label

Fields: rx_bytes(bytes received)(float), tx_bytes(bytes transmitted)(float), rx_errors(integer), tx_errors(integer), link_up(integer), connected_clients(integer), utilization_pct(float), latency_ms(float)

4. Source - SNMP/RouterOS, Interval - 60s, 3 fields

Measurement: router_metrics

Tags: device_id, device_name, device_type

Fields: cpu_percent(float), mem_percent(float), uptime_seconds(float)

5. Source - SNMP/NUT, Interval - 60s, 8 fields

Measurement: ups_metrics

Tags: device_id, device_name, location, communication_type

Fields: battery_charge_pct(float), runtime_remaining_min(float), load_pct(float), input_voltage(float), output_voltage(float), on_battery(boolean), battery_voltage(float), temperature (ups internal temperature if reported by device)(float)